

Summary of Agent Framework: Multi-agent Systems

Rui Zhou (ruizhou3), Sudhit Rao (srrao2), Vir Sikand (vsikand2)

Problem and Motivation

Autogen:

As LLMs have been tasked with handling more complex requests and queries with the advancement of their size and complexity, there have been increasing demands in regard to their reasoning ability and knowledge. Single Agents are not able to fully utilize an entire skillset and a more diverse array may be necessary. As such, there is an absence of a general framework for handling multi-agent coordination and cooperation. The collaboration of multiple agents allows for feedback loops and increased performance. In addition, a multi-agent framework makes it easier to handle these types of interactions in order to provide better responses and behavior for the user.

CAMEL:

There are various different types of LLMs these days, but all of them need heavy human involvement when solving complex problems. Although models like GPT-4 might already be very powerful, they still have this problem because they lack the ability of effective prompt generation. As a result, the authors design a framework that can help reduce the human involvement. Specifically, they try to assign different tasks to different agents to let them automatically cooperate to solve a complex problem.

Related Work

Autogen

One of the examples of related works is CAMEL, this is fixed on a two-agent structure, which is a more strict requirement. Autogen allows for any number. There is also ChatDev, which simulates multiple developer roles in an organization. Autogen allows us to establish any sort of behavior to the agents. Another example is MetaGPT which uses multiple agents to structure the workflow of agents with specified instructions.

Multi-agent debate framework

This framework is designed to have multiple agents that can debate with each other. To be specific, unlike only using one agent to help answer the question and hoping it answers correctly, the framework assigns the same task to different agents and all the agents will have their own answer. After the answer generation, all the agents will be involved in a debate in which they critique each other's answers over a few rounds to get a common answer for all the agents. In this way, the final answer will be very reasonable and reliable.

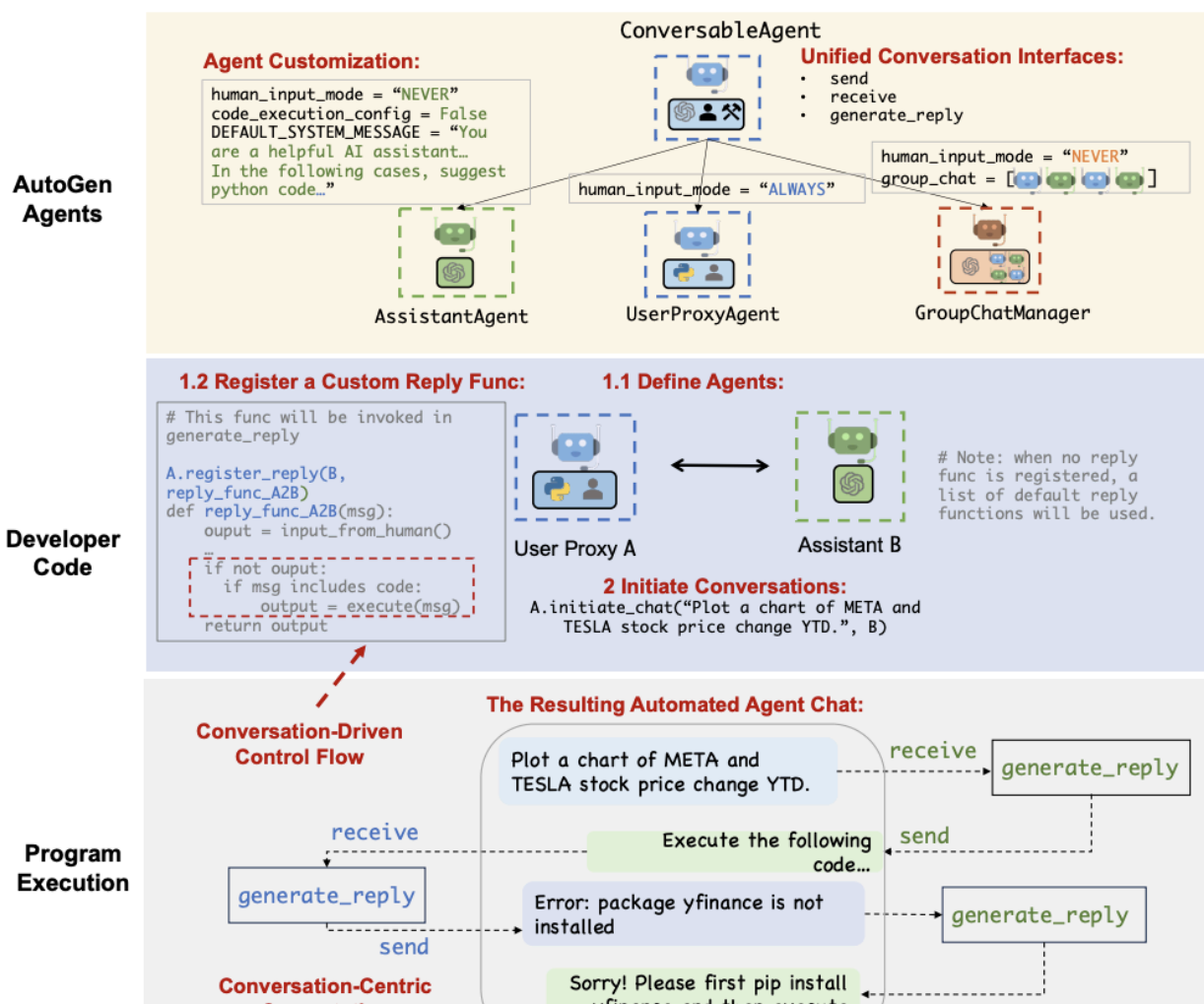
Solution Overview

Autogen

Autogen helps developers build multi-agent LLM applications. The main way it streamlines the development and information process is through conversations between multiple agents. The main concepts are conversable agents and conversation programming.

Conversable agents send and respond to messages to each other by taking turns. They have their own internal context based on the conversations. In addition, they have various defined abilities based on their backend from tools or a large language model. There are also built-in types like an assistant and they can be custom as well.

Conversation programming is the second paradigm involved. These define how the agents interact with one another. This can be done through LLM calls, as well as human input. There are a number of key functions involved related to sending, replying, and generating information. The conversation continues until there is a termination



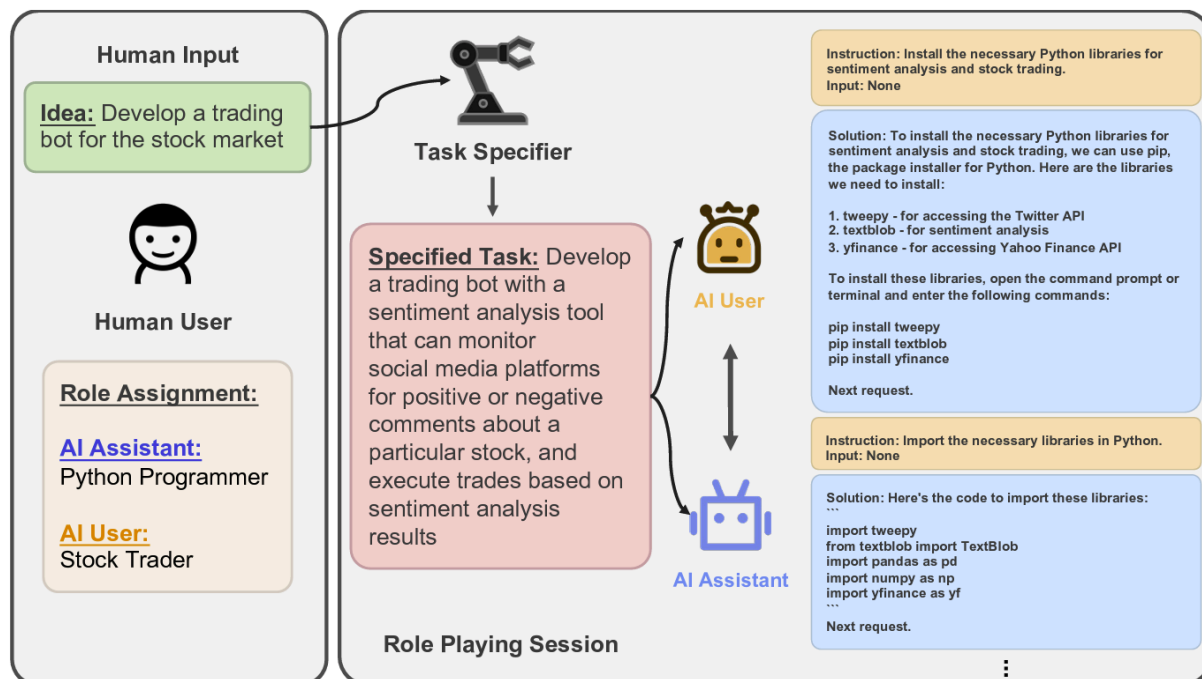
requirement that occurs. There are also language prompts that are used in conjunction with programming languages.

CAMEL

The framework is a novel role-playing approach that starts with the human. The human user will start with a simple question or an idea about what he really wants to do. There will be two AI agents who will mostly help solve the problem. One is an AI assistant agent that helps write the code to try to solve the problem. The other one is an AI user who works mainly like the human that gives instructions to the AI agents after each step.

Between the human user and the AI agents, there will be a Task Specifier that helps make the human ideas more specific and actionable. Then, the task manager will give the task to the AI user and AI assistant. In this way, it is easier for the AI agents to solve the task.

Moreover, to make sure the conversation between each task is reasonable, the author introduces a critic agent that can help select answers and provide feedback for each agent. When encountering problems like infinite loop of messages (e.g. the two agents keep saying “thank you” and “goodbye” forever), the critic agent can help stop this loop.



Limitations

CAMEL [1]

There are four challenges observed during the framework development.

- Role Flipping: The AI agent and the AI assistant flip their own roles during the process.
- Assistant Repeats Instruction: The assistant simply repeats the user's instructions.
- Flake Replies: the assistant only replies, "I will..." which is a specific solution.
- Infinite Loop of Messages: The assistant and user engage in an infinite loop conversation.

The author tried to solve this problem by using the critic agent but they still think it is challenging to completely solve the problem. Also, there are potential ethical problems. The paper highlighted "Bad Mind", which means the agents collaborate on harmful tasks.

Autogen [3]

- Complexity in Workflow Design: Designing optimal multi-agent workflows involves many decisions (e.g., number of agents, roles, interaction patterns), and there is no one-size-fits-all solution. This adds complexity to application development.
- Scalability and Debugging Challenges: As the number of agents and interactions increases, it becomes harder to track, debug, and manage the system's behavior, potentially leading to unintelligible or inefficient agent interactions.

Future Research Directions

CAMEL [1]:

More advanced critic agents can be developed to help with the answer generation. Hopefully, it can help solve the four limitations completely.

Autogen [3]:

- Integrating existing agent implementations into their multi-agent framework

- Investigate different agent topology strategies to improve conversational behavior and determine which topology and framework is best for a certain type of task
- Implementing robust safety protocols to manage risks associated with fully autonomous multi-agent interactions
- Seamless tool integration with external frameworks

Summary of Class Discussion

Autogen

Q1: In a RAG setup, how many attempts does AutoGen make before terminating? Should this be configurable?

A1: This might not be specified, but it could be configurable in order to improve the application

Q2: How would you see humans involved with the multi-agent process as it scales in the real world?

A2: It definitely could scale and have applications in the real world. As it gets larger, it could be more difficult.

Q3: What are the ethical concerns involved with allowing multiple autonomous agents to make decisions on their own?

A3: There are ethical concerns without having restrictions on the actions the agents can perform.

CAMEL

Q1: What is the task specifier that is not correct?

A1: We can use the more advanced model to help solve this problem. Also, we can train our task specifiers to make it more specific and actionable.

Q1: For different agents, will a more advanced model help?

A1: Yes. The model's performance might be better if a better model is used.

Reference

[1] Li, G., Hammoud, H., Itani, H., Khizbullin, D., & Ghanem, B. (2023). Camel: Communicative agents for "mind" exploration of large language model society. *Advances in Neural Information Processing Systems*, 36, 51991-52008.

[2] Du, Y., Li, S., Torralba, A., Tenenbaum, J. B., & Mordatch, I. (2023). Improving factuality and reasoning in language models through multiagent debate. In *Forty-first International Conference on Machine Learning*.

[3] Wu, Qingyun, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W White, Doug Burger, Chi Wang. "AutoGen: Enabling Next-Gen LLM Applications via Multi-Agent Conversation." arXiv:2308.08155, 2023.